

Video 2: MECH5200 Numerical Methods for PDEs : *Introduction to MATLAB and Matrices*

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Recall the example where we scaled 2 Independent 2D Vectors

- The addition of scaled vectors:

$$\text{NewVector} = \alpha \cdot \begin{bmatrix} 2 \\ 3 \end{bmatrix} + \beta \cdot \begin{bmatrix} 1 \\ 6 \end{bmatrix}$$

- This can be written using a more compact notation (Matrix-Vector-Product) as:

$$\begin{bmatrix} 2 & 1 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \quad (1)$$

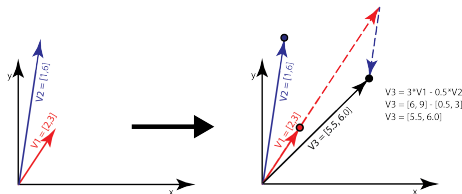


Figure: Column vectors to column space of a Matrix.

Matrices and Multiplication

- First, what is a matrix? Two dimensional array, with entries defined using row, column indexing:

$$A = \begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \\ a_{3,1} & a_{3,2} & a_{3,3} \end{bmatrix} \quad (2)$$

- In MATLAB: $A = [1, 2, 3; 4, 5, 6; 7, 8, 9];$
 - The comma separates entries into columns, the semi colon separates entries into independent rows.
- We can think of a matrix as row vectors and/or column vectors.

Matrices and Multiplication

- Like vectors, matrix entries can be accessed individually in MATLAB using:

$$a(i,j) = a_{i,j} \quad (3)$$

- For example, if:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \quad (4)$$

- Try this:** $A(1,2) = ?$
- Try this:** Loop over the entries and add the product of the row and column location.
 - e.g.: $A(2,3) = A(2,3) + 2*3;$

Matrices and Multiplication

- In this course, we will see linear systems of equations that need to be solved:

$$a_{11}x + a_{12}y + a_{13}z = b_1$$

$$a_{21}x + a_{22}y + a_{23}z = b_2$$

$$a_{31}x + a_{32}y + a_{33}z = b_3$$

- These equations can be expressed in matrix form:

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \quad (5)$$

- Which will further simplify to:

$$[A]x = b \quad (6)$$

- Often, we are interested in finding x given $[A]$ and b .
- But....first, let's understand matrix-vector multiplication.

Multiplication of a matrix and a vector (MVP)

- Matrix Multiplication = the *scaling of column vectors of A by the entries of x*:

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} a_{11} \\ a_{21} \\ a_{31} \end{bmatrix} x_1 + \begin{bmatrix} a_{12} \\ a_{22} \\ a_{32} \end{bmatrix} x_2 + \begin{bmatrix} a_{13} \\ a_{23} \\ a_{33} \end{bmatrix} x_3 = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

- The properties of the column space of matrix A is important in MVP.
- If the columns of A ($N \times N$ matrix) are not independent, then the matrix has a limited ability to create RHS vectors (Matrix is singular).

Example: Multiplication of a matrix and a vector (MVP)

- Matrix Multiplication = the *scaling of column vectors of A by the entries of x*.
- Here A = the identity matrix, I , so we are scaling the unit vectors!:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1.5 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} 2 + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} 1.5 + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} 3 = \begin{bmatrix} 2.0 \\ 1.5 \\ 3.0 \end{bmatrix}$$

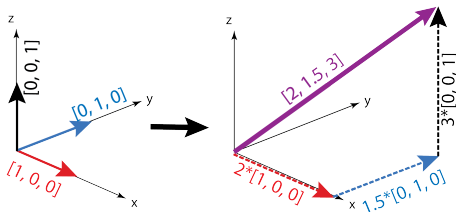


Figure: Column vectors to column space of a Matrix.

MATLAB and Matrices

Let's try to code a matrix-vector product in MATLAB with this new approach – and visualize this form of multiplication:

- Get the size of the matrix ($N = \text{length}(A(:,1))$)
- Initialize the "answer" vector ($b = \text{zeros}(N,1)$)
- Cycle over i columns of the matrix
 - Scale the i -th column of the matrix by $x(i)$
 - Add the result to the "answer" vector b .
- Check the answer.

Let's give this a try!!!

Multiplication of Vectors: Column vector scaling

MATLAB code

```
A = [1 2 3; 4 5 6; 7 8 9];  
x = [1; 2; 3];  
b = [0; 0; 0];  
  
% Performing A*x as a sum of scaled columns  
for(j=1:3)  
    b = b + A(:,j)*x(j);  
end
```

Matrices and Multiplication – Row-Vector Dot Products

We can also interpret the Matrix-Vector Product (MVP) as a series of Matrix-row and column vector dot products:

$$\begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \\ a_{3,1} & a_{3,2} & a_{3,3} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \quad (7)$$

When multiplied out:

$$\begin{bmatrix} a_{1,1} * x_1 + a_{1,2} * x_2 + a_{1,3} * x_3 \\ a_{2,1} * x_1 + a_{2,2} * x_2 + a_{2,3} * x_3 \\ a_{3,1} * x_1 + a_{3,2} * x_2 + a_{3,3} * x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \quad (8)$$

MATLAB and Matrices

Let's try to code matrix-vector multiplication in MATLAB:

- Cycle over i rows of the matrix
 - Zero the i -th answer ($b(i) = 0$)
 - Cycle over j columns of the matrix.
 - Multiply the i, j -th entry of the matrix with the j -th entry of the x -vector
 - Add the result to the i -th answer ($b(i) = b(i) + A(i, j) * x(j)$)

Let's give this a try!!!

Matrix-Vector-Product – summary

Column Vector scaling version of the MVP:

```
A = [1 2 3; 4 5 6; 7 8 9]; x = [1 ; 2; 3];  
for(i = 1:3)  
    b(i) = 0;  
    for(j = 1:3)  
        b(i) = b(i) + A(i,j)*x(j);  
    end  
end
```

Dot Product version of the MVP:

```
A = [1 2 3; 4 5 6; 7 8 9]; x = [1 ; 2; 3];  
b = [ 0  0  0];  
for(i = 1:3)  
    for(j = 1:3)  
        b(i) = b(i) + A(i,j)*x(j);  
    end  
end
```

MATLAB and Matrices

- Matrix-vector multiplication in MATLAB:

$$\textit{Result} = A * \textit{Vec} \quad (9)$$

Matrices and Multiplication with Vectors!

Great! Now that we can multiply $A * x = b$, what does this multiplication *actually* represent?

$$\begin{bmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \quad (10)$$

Activity: Determine what the matrix 'A' does to 'b' to make it equal 'x'.
Pause the video and try to rationalize what this means.

Matrix-Vector Product: Recall the example where we scaled 2 Independent 2D Vectors

- The pair of (independent) column vectors of A can be scaled and added to produce a resultant vector in the 2D space.

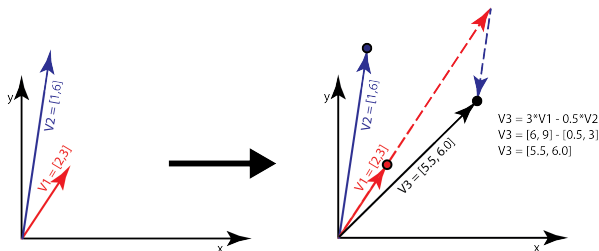


Figure: The addition of two independent vectors (scaled) produces a third vector in the space.

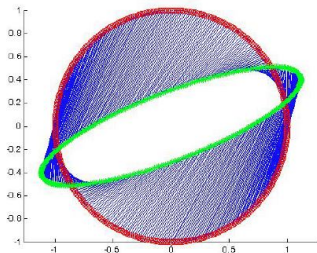
- For an $N \times N$ matrix, if the column vectors are not independent, the Matrix is called singular (the column vectors can not reach the N -dimensional space).

Matrices and Multiplication

One way to understand matrix multiplication in two-dimensions is to:

- Consider the unit vectors that make a unit circle around the origin:
 - U_i : Unit vector, defined such that $\|U_i\| = 1$.
- Multiply each of these unit vectors by the matrix A and plot the resulting vector b

Example result:



Matrices and Multiplication

MATLAB code for previous example:

```
%===== Matrix as a transformation =====  
close all; figure; hold on;  
%=== Declare the 2x2 matrix that we are looking at  
A = [1 -.5; .5 .1];  
cond(A)  
%=== calculate the points on the unit circle theta = 1:360;  
x = cos((theta)*pi/180);  
y = sin((theta)*pi/180);  
%=== calculate the transformation due to the matrix  
for(i= theta)  
X = [x(i); y(i)];  
trans = A*X;  
xx(i) = trans(1);  
yy(i) = trans(2);  
plot([x(i),xx(i)],[y(i),yy(i)])  
end  
plot(x,y,'ro')  
plot(xx,yy,'g*')  
axis equal
```

Matrices and Multiplication

This shows us that:

- A matrix will change magnitude and direction of a vector
- A matrix will tend to transform the unit circle into an ellipse
- Depending on entries, transformation can be more or less extreme.
- Axes of the ellipse tell us something about how easy it is to solve linear system.

$$A * x = b \quad (11)$$

- In most numerical methods we know A and b and we are interesting in finding x .

$$x = A^{-1}b \quad (12)$$

- Therefore, we are interested in the inverse mapping (think about this as: from the ellipse to the circle)
- The closer the mapping A is to a circle, the easier this will be.

Matrices : Condition Number

- Condition number (in MATLAB: $\text{cond}(A)$):
- Represents how well conditioned a matrix is.
 - A large condition number : ill-conditioned system
 - A small condition number : well-conditioned system
- The condition number is the ratio of the singular values of the matrix
- Type 'help SVD' in MATLAB or $SVD(A)$.
- Condition number tells us how sensitive the solution (mapping output) is to a perturbation in the RHS vector.

Some Useful Factorizations of a Matrix (we may use some of these)

- Factorizing $[A]$ into upper $[U]$ and lower $[L]$ triangular matrices ($[L]$ has ones on the diagonal).
 - This is a convenient factorization for repeated matrix solution
 - Direct method: a version of this is used by default in MATLAB item In MATLAB : $[L,U] = \text{lu}(A)$

$$[A]x = [L][U]x = b \rightarrow [U]x = [L]^{-1}b \quad (13)$$

- Factorize $[A]$ into matrices $[Q]$ and $[R]$ (Orthogonal-triangular decomposition):
 - Matrix $[Q]$ has orthogonal unit vectors as columns
 - Matrix $[R]$ is an upper triangular matrix (numbers on diagonal and above diagonal)
 - In MATLAB: $[Q,R] = \text{qr}(A)$

$$[A]x = [Q][R]x = b \rightarrow [R]x = [Q]^T b \quad (14)$$

Some Useful Factorizations of a Matrix (we may use some of these)

- Eigen-decomposition of a matrix $[A]$:

- Matrix $[V]$: Columns of $[V]$ are eigenvectors of $[A]$
- Matrix $[D]$: diagonals of $[D]$ are eigenvalues of $[A]$
- High relevance for systems of ODEs (and DE's)
- see MATLAB: $[V,D] = \text{eig}(A)$:

$$[A] = [V][D][V]^{-1} \rightarrow [A][V] = [V][D] \quad (15)$$

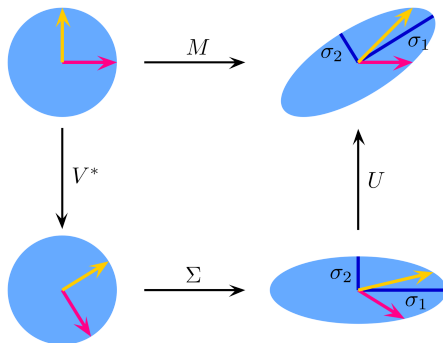
- Singular value decomposition (SVD) of a matrix $[A]$

- Matrices: $[U]$ and $[V]$: orthonormal unitary/singular vectors
- Matrix: $[\Sigma]$ diagonals are the positive singular values of $[A]$
- see MATLAB $[U,S,V] = \text{svd}(A)$

$$[A] = [U][\Sigma][V]^T \rightarrow [A][V] = [U][\Sigma] \quad (16)$$

The singular value decomposition

- The SVD can be thought of as a sequence of transformations. The SVD page from wikipedia shows a nice diagram of this:



$$M = U \cdot \Sigma \cdot V^*$$

https://en.wikipedia.org/wiki/Singular_value_decomposition

- The $[V]^T$ rotates the vectors, followed by the stretch by $[\Sigma]$, then another rotation $[U]$.

Summary: What have we learned?

- A matrix is a N-row $\times$ M-column rectangular array of numbers (also is a 2D array of numbers).
- Row-column indexing matrices is relatively easy using MATLAB syntax: 'A(i,j)' to access A_{ij} .
- A matrix-vector product (matrix multiplied by a vector) is common in numerical methods: $Ax = b$
- A matrix can be "seen" as a linear transformation – by performing an MVP, the matrix transforms the magnitude and direction of a vector.
- The matrix transformation (circle $\rightarrow$ ellipse) is an indication of the condition number ($M = \text{cond}(A)$) of a matrix.
- Matrices can be factored into several common (and useful) decompositions. See linear algebra.